Local actions, invariant pencils and scalar quotients

Human Compression project: private unreviewed draft

This common paper develops the local-action arguments used in a complete corrected recovery of Buekenhout–Dehaye–Leemans [1]. The complete application is written separately and remains subject to distinct review. Large classification inputs will remain explicitly attributed; none is inferred from the small calculations accompanying this draft.

1 Local actions and preserved partitions

A finite linear space has a unique line through two distinct points, at least two points on each line, and at least two lines on each point. The specified faithful group G is part of the data. Its pencil at x is the set $\mathcal{L}(x)$ of lines through x. We use local primitivity for primitivity of both G_x on $\mathcal{L}(x)$ and G_L on the points of L. For a flag-transitive linear space this is equivalent to the source's residual weak primitivity: the remaining global weak-primitivity condition follows from the retained Higman–McLaughlin point-primitivity theorem. For translation structures below it also follows directly from the proof.

Lemma 1.1 (Supergroups and coarsening)**.** *A permutation group containing a primitive subgroup is primitive, and one containing a two-transitive subgroup is two-transitive. Suppose a group H acts transitively on a finite set X and preserves a partition of X with at least two parts, each of size at least two. Then its action is not primitive. In particular, this conclusion applies to a pencil partition induced by a genuinely H-invariant coarser incidence structure.*

Proof. A block system for the larger group is one for the subgroup; transitivity of the subgroup rules out extra orbits. Transitivity on ordered distinct pairs also passes to a supergroup. The partition in the second assertion is, by hypothesis, a nontrivial invariant equivalence relation, which is exactly an obstruction to primitivity. Notice that ordering two line sizes alone does not construct such a relation. □

Proposition 1.2 (Translation local actions)**.** *Let V be a finite vector space over $\mathbb{F}_p$, let $\mathcal{S}$ be a spread of nonzero proper additive subspaces, and let $H \leq \mathrm{GL}(V)$ preserve $\mathcal{S}$ and act transitively on it. The cosets of members of $\mathcal{S}$ are the lines of a translation linear space. Put $G = V \rtimes H$. Then G is flag-transitive. It is locally primitive exactly when H is primitive on $\mathcal{S}$ and H_U acts irreducibly on U for a member $U \in \mathcal{S}$. It is locally two-transitive exactly when H is two-transitive on $\mathcal{S}$ and H_U is transitive on $U \setminus \{0\}$. In the primitive case G is point-primitive as well.*

Proof. Translations are transitive on points and move each affine pencil to the pencil at zero. The latter pencil is $\mathcal{S}$, with induced group H. The setwise stabilizer of the line U induces $U \rtimes H_U|_U$ on it. For a regular translation group, a block through zero is an additive subgroup: if b lies in the block, translation by b makes the block meet itself and hence fixes it setwise. Conversely every invariant additive subgroup gives its coset block system. Thus the affine line action is primitive exactly when its linear stabilizer is irreducible. It is two-transitive exactly when the zero stabilizer is transitive on nonzero vectors. This proves both equivalences, including possible kernels of the restriction $H_U \to \mathrm{GL}(U)$.

Finally let W be a nonzero H-invariant additive subspace of V. Choose $0 \neq w \in W$ and the unique spread member U containing it. The intersection $W \cap U$ is H_U-invariant, so irreducibility gives $U \subseteq W$. Transitivity on the spread gives every member inside W, and these members cover $V \setminus \{0\}$. Hence $W = V$, proving point-primitivity without a separate global classification. □

Lemma 1.3 (A field detected by a commutant)**.** *Suppose $N \leq \mathrm{GL}_{\mathbb{F}_p}(V)$ and its commuting algebra is a field $Q = \mathrm{End}_{\mathbb{F}_p N}(V)$ acting as scalars on V. Every linear transformation normalizing N is semilinear on the Q-space V. If $\mathbb{F}_q \subsetneq Q$, $\dim_Q V \geq 2$, and H normalizes N, then H is imprimitive on the $\mathbb{F}_q$-lines whenever it acts on those lines: the Q-lines give a nontrivial invariant partition.*

Proof. For $h \in N_{\mathrm{GL}_{\mathbb{F}_p}(V)}(N)$, conjugation sends the commuting algebra of N onto itself. Its restriction to Q is a field automorphism σ_h, and $h(av) = \sigma_h(a)h(v)$. Thus h permutes Q-lines. Each Q-line contains exactly $(|Q| - 1)/(q - 1) > 1$ distinct $\mathbb{F}_q$-lines, and there is more than one Q-line. Different Q-lines are disjoint away from zero. Lemma 1.1 applies. □

For example, for the natural module of $\mathrm{SL}_m(Q)$, $m \geq 2$, the commuting algebra over $\mathbb{F}_p$ is exactly Q. Indeed commuting with all $I + tE_{ij}$ first forces the same additive map on every coordinate and then forces that map to commute with multiplication by every $t \in Q$. It is multiplication by a scalar. This supplies the actual invariant partition in the source's extension-field special-linear branch. The corresponding symplectic and exceptional module identifications must be stated at their precise retained scope; the assertion is not justified by an abstract subgroup name alone.

Proposition 1.4 (Circles)**.** *Let Ω have at least three points and let its lines be all two-element subsets. A faithful group G is flag-transitive on this linear space if and only if it is two-transitive on Ω. In that case it is residually weakly primitive if and only if G_x is primitive on $\Omega \setminus \{x\}$, and locally two-transitive if and only if G is three-transitive. If G also preserves a thick linear space on the same point set, its circle action is not residually weakly primitive.*

Proof. Flags $(x, \{x, y\})$ are ordered pairs of distinct points. The pencil at x identifies with $\Omega \setminus \{x\}$. A line stabilizer interchanges its two points by two-transitivity, so its two-point action is already primitive and two-transitive. Global point-primitivity follows from two-transitivity. This proves both equivalences.

In the final assertion the sets $L \setminus \{x\}$, for thick-space lines L through x, partition $\Omega \setminus \{x\}$. Every part has at least two elements and there are at least two parts. They form an invariant partition for G_x, so Lemma 1.1 applies. In particular the word "thick" cannot be omitted when using this test on $\mathrm{AG}(m, q)$: for $q = 2$ that space is itself the circle. □

2 Classical pencils and the small affine exclusions

Proposition 2.1 (Classical local actions)**.** *The natural groups containing $\mathrm{PSL}_{m+1}(q)$ on $\mathrm{PG}(m, q)$, or containing translations and $\mathrm{SL}_m(q)$ on $\mathrm{AG}(m, q)$, are flag-transitive and locally two-transitive for $m \geq 2$. In the affine plane the induced special-linear pencil group is $\mathrm{PSL}_2(q)$; it need not contain $\mathrm{PGL}_2(q)$.*

Proof. Special-linear matrices are transitive on nonzero vectors and on ordered pairs of distinct one-dimensional subspaces: choose independent representatives, extend them to bases, and rescale a representative or a complementary vector to correct the determinant. In the projective pencil at a point, any quotient matrix A extends to the determinant-one matrix $\mathrm{diag}((\det A)^{-1}, A)$. On a projective line, any $\mathrm{GL}_2(q)$ action extends by its inverse determinant on a complementary coordinate. Both local projective actions are therefore two-transitive. The affine pencil is precisely the projective action on vector lines. Its vector-line stabilizer contains every nonzero multiplier on that line, compensating the determinant on a complement; line translations give two-transitivity there. Supergroups preserve these properties.

For $m = 2$ the scalar kernel in $\mathrm{SL}_2(q)$ has order $\gcd(2, q - 1)$, giving exactly $\mathrm{PSL}_2(q)$ on the pencil. For odd q this has index two in $\mathrm{PGL}_2(q)$. Thus $\mathrm{ASL}_2(3)$ on $\mathrm{AG}(2, 3)$ has pencil group A_4, not S_4, despite being locally two-transitive. The $\mathrm{PGL}_2(q)$ inclusions in source Propositions 3.7 and 3.8 must be replaced by $\mathrm{PSL}_2(q)$. The source's final special-linear branches do not require this erroneous inclusion. □

Proposition 2.2 (Natural symplectic pencils)**.** *For even $m \geq 4$, the natural $\mathrm{Sp}_m(q)$ is primitive but not two-transitive on vector lines. Its normalizer over $\mathbb{F}_p$ preserves perpendicularity, up to field automorphism and nonzero form multiplier. Consequently every group between it and its normalizer has a locally primitive, but not locally two-transitive, affine-space action.*

Proof. Symplectic basis extension shows that the stabilizer of a vector line P has three orbits on vector lines: P, the remaining points of $P^\perp$, and those outside $P^\perp$. Write

$$v = (q^m - 1)/(q - 1), \qquad a = (q^{m-1} - 1)/(q - 1).$$

A block through P is a union of stabilizer orbits, so its possible proper nontrivial sizes are a and $q^{m-1} + 1$. But $\gcd(a, v) = 1$, and $v/2 < q^{m-1} + 1 < v$; neither size divides v. This proves primitivity, and the two nonempty other orbits disprove two-transitivity.

The commuting algebra over $\mathbb{F}_p$ is $\mathbb{F}_q$. Indeed commuting with every transvection $x \mapsto x + t\langle x, w\rangle w$ forces an endomorphism to preserve each $\mathbb{F}_q w$, the image after subtracting the identity. An additive map preserving all vector lines is a common scalar: apply it to $u, v, u + v, u + cv$ for independent u, v and $c \in \mathbb{F}_q$. Lemma 1.3 now makes a normalizer element semilinear over $\mathbb{F}_q$. After removing its field automorphism, pulling back the alternating form gives another invariant form. Relative to the original nondegenerate form its defining operator commutes with the group, and hence is a scalar. Perpendicularity is preserved even by the full normalizer. Finally the symplectic group is transitive on nonzero vectors, so a vector-line stabilizer is transitive on the nonzero vectors of that line. Proposition 1.2 gives the remaining local assertions. □

Lemma 2.3 (A small normal subgroup test)**.** *A nontrivial normal subgroup in a faithful primitive permutation group is transitive. In particular its order is at least the degree. A faithful primitive group with a nontrivial center is regular of prime order. These tests apply to the induced group after removal of the restriction kernel.*

Proof. Normal-subgroup orbits are a block system. All singleton orbits would make the subgroup trivial by faithfulness. A central transitive subgroup is regular, since an element fixing one point fixes all. Its centralizer in the symmetric group is its own regular copy, forcing the whole group to be abelian regular. Primitivity of a regular group is equivalent to having no proper nontrivial subgroup, hence to prime order. □

Retain Huybrechts' actual representation cases as quoted in source Proposition 3.4, rather than just abstract isomorphic subgroup names. The exceptional degree-364 pencil group is $\mathrm{PSL}_2(13)$, of order 1092. A point stabilizer has order three and lies properly in a cyclic split torus of order six: an order-three element diagonalizes over $\mathbb{F}_{13}$, and the determinant-one diagonal torus modulo scalars has order six. Thus this action is imprimitive. In the degree-40 cases $2^4.A$, the normal subgroup of order sixteen has a nontrivial image of order at most sixteen on directions: the scalar kernel in the linear $\mathbb{F}_3$-action has order at most two. Such an image cannot be transitive of degree forty, so Lemma 2.3 excludes primitivity. In the remaining $4.A_5$ or $N.S_5$ cases ($|N| \leq 4$), the specified normal subgroup of order at most four likewise has either a nontrivial, nontransitive image, or lies entirely in the restriction kernel. In the latter case the induced group is a quotient of A_5 or S_5. The former order is not divisible by forty. A transitive S_5 quotient of degree forty must be S_5 itself (its other proper quotients have order at most two); its order-three point stabilizer lies properly in an S_3. Thus this action is imprimitive as well. This supplies the missing explicit local exclusions, conditional on the retained representation classification.

Proposition 2.4 (The exceptional affine plane)**.** *Retain Foulser's ten flag-transitive actions on $\mathrm{AG}(2, 9)$ with induced pencil group A_5 or S_5, each containing the specified minimal $F = 3^4 \rtimes 2.A_5$ in its natural $\mathrm{SL}_2(9)$ representation. All ten are locally primitive and none is locally two-transitive.*

Proof. In the degree-ten action A_5 has point stabilizer S_3, which is maximal. Indeed a proper overgroup could only have order twelve or thirty; an order-twelve subgroup is A_4 and contains

no S_3, and simplicity forbids index two. Neither 60 nor 120 is divisible by $10 \cdot 9$, excluding two-transitivity for either pencil image.

Choose an involution of A_5 and a lift f in $2.A_5$. The only involution in $\mathrm{SL}_2(9)$ is $-I$: a determinant-one involution in odd characteristic is diagonalizable with both eigenvalues -1, unless it is the identity. Thus f has order four and $f^2 = -I$. The polynomial $X^2 + 1$ has two distinct roots in $\mathbb{F}_9$, so f fixes two vector lines. On either line its scalar of order four generates $\mathbb{F}_9$ over $\mathbb{F}_3$. It acts irreducibly on the additive two-dimensional line. With line translations this gives a primitive line action; flag transitivity transfers it to every line. Supergroups preserve both local primitivity conditions.

Only the cyclic subgroup generated by f is required to fix a direction. The whole Sylow subgroup need not do so. This avoids the stronger simultaneous Sylow-fixed-point assertion in the printed proof of Proposition 3.9 while supplying its claimed conclusion. □

For the other Foulser plane cases, a faithful primitive action of A_4 or S_4 has degree at most four, by the normal Klein four subgroup. The elementary maximal-subgroup list A_4, D_{10}, S_3 for A_5 gives primitive degrees $5, 6, 10$. Comparing the specified $q + 1$ values in source Proposition 3.7 leaves only $q = 9$ among its exceptional cases.

3 Semilinear spread rigidity and the one-dimensional branch

The following full argument is reused from the reviewed phase4 theorem *Rigidity of locally primitive semilinear spreads*. Its application to the source question is already recorded there; this edition claims no new question answer or priority. We retain Delandtsheer's reduction quoted as source Proposition 5.2: a one-dimensional flag-transitive action contains all translations, and its spread case consists of cosets of equal-dimensional additive subspaces. The proof below includes the restriction kernels and characteristic-dividing Galois case.

Lemma 3.1 (Prime degree and arithmetic)**.** *A finite primitive permutation group with cyclic normal subgroup and cyclic quotient has prime degree, if its degree exceeds one. If*

$$r = \frac{p^{u\ell} - 1}{p^{\ell} - 1}$$

is prime, where p is prime and $u > 1$, then u is prime, $\ell = u^e$ for some $e \geq 0$, and $p \not\equiv 1 \pmod{u}$.

Proof. Pass to the faithful induced action. The cyclic normal subgroup either is trivial or is transitive. In the former case the whole induced group is cyclic, hence regular of prime order. In the latter case the cyclic normal subgroup is regular. Any proper nontrivial characteristic subgroup would give a nontrivial block system preserved by the whole group, so its order and the degree are prime.

If u is composite, its geometric sum factors into two integer geometric sums greater than one. Thus u is prime. If a prime $v \neq u$ divides ℓ, put $x = p^{\ell/v}$. Reduction modulo $\Phi_u(X) = 1 + X + \cdots + X^{u-1}$ shows that $\Phi_u(X)$ divides $\Phi_u(X^v)$: multiplication by v permutes the nonzero residues modulo u. It follows that $1 < \Phi_u(x) < \Phi_u(x^v) = r$, contradicting primality. Hence $\ell = u^e$. Finally $p \equiv 1 \pmod{u}$ would give $u \mid r$ while $r > u$, again impossible. □

Lemma 3.2 (Cyclic semilinear descent)**.** *Let p, u be primes with $p \not\equiv 1 \pmod{u}$. Let a finite group J act faithfully and irreducibly on an $\mathbb{F}_p$-space U of dimension u^e. Suppose $C \triangleleft J$ is cyclic, $K = \mathbb{F}_p[C]$ is a field of endomorphisms, J normalizes K, and J/C is a cyclic u-group. Then $\dim_K U = 1$.*

Proof. Write $|K| = p^h$. Since $h \mid u^e$, h is a power of u. For every power u^a, the number $p^{u^a} - 1$ is prime to u: this is immediate for $p = u$, and otherwise follows from $p^{u^a} \equiv p \not\equiv 1 \pmod{u}$. In particular C has order prime to u.

Choose a lift t of a generator of J/C and let y be its u-primary component inside the cyclic group generated by t. The prime-to-u component lies in C, so $J = C\langle y\rangle$. Let σ be the automorphism of K induced by y and let m be its order. Then $z = y^m$ centralizes K, C and y, hence J. Every nonzero element of the commutative algebra $\mathbb{F}_p[z]$ is invertible: its kernel is J-invariant, so irreducibility makes it injective. This finite algebra is a field of order p^a, with $a \mid u^e$. Its multiplicative group has order prime to u. Since z has u-power order, $z = 1$. Thus the order of y is exactly m.

There is a nonzero y-fixed vector without dividing by m. Choose $0 \neq v \in U$ and consider

$$S(b) = \sum_{i=0}^{m-1} \sigma^i(b) y^i(v), \qquad b \in K.$$

Every $S(b)$ is fixed by y, by cyclically shifting the summands. They cannot all vanish. Apply a K-linear functional nonzero on v; vanishing would give a nonzero linear relation over K among the distinct field automorphisms $1, \sigma, \ldots, \sigma^{m-1}$. Their linear independence follows by taking a shortest relation and subtracting its value at bx from a suitable multiple of its value at x, shortening the relation. For some b we therefore obtain a nonzero fixed vector $w = S(b)$. Its K-line is J-invariant: C acts by scalars and y normalizes K while fixing w. Irreducibility gives $U = Kw$. □

Theorem 3.3 (Locally primitive one-dimensional spread spaces)**.** *Let $F = \mathbb{F}_{p^d}$, let $\mathcal{S}$ be a spread of ℓ-dimensional additive subspaces, with $0 < \ell < d$, and let $H \leq \Gamma L_1(F)$ preserve it and act transitively. Suppose H is primitive on $\mathcal{S}$ and H_U is irreducible on a spread member U. Then $\mathcal{S}$ is exactly the set of one-dimensional $\mathbb{F}_{p^\ell}$-subspaces of F. Its size is prime and $d = u^{e+1}, \ell = u^e$ for a prime u. Conversely such a locally primitive one-dimensional action exists precisely when $(p^d-1)/(p^\ell-1)$ is prime; the geometry is then uniquely $\mathrm{AG}(d/\ell, p^\ell)$ up to isomorphism. This does not classify every possible acting subgroup.*

Proof. The normal scalar subgroup of H is cyclic and its quotient embeds in the cyclic Galois group. Lemma 3.1 gives prime spread size $r = (p^d-1)/(p^\ell-1)$. Divisibility of the denominator into the numerator implies $\ell \mid d$: dividing $d = a\ell + b$, $0 \leq b < \ell$, would otherwise give $p^\ell - 1 \mid p^b - 1$. Apply the arithmetic part of that lemma with $u = d/\ell$.

Put $C_0 = H_U \cap F^*$, and let J be the faithful image of H_U on U. The ambient scalar group C_0 restricts faithfully on nonzero U, and $K = \mathbb{F}_p[C_0]$ is an actual subfield of F. Also U is a K-space. The semilinear stabilizer normalizes K; its faithful image modulo the image C of C_0 is a quotient of a subgroup of $\mathrm{Gal}(F/\mathbb{F}_p)$, hence a cyclic u-group because $d = u^{e+1}$. Lemma 3.2 now gives $U = wK$ with $|K| = p^\ell$. The unique subfield of that order is $\mathbb{F}_{p^\ell}$. All H-images of U are K-lines, since scalar-Frobenius maps preserve this subfield. There are exactly r such lines, so these are the entire spread.

For the converse use $F^+ \rtimes F^*$ on the K-line spread. It is flag-transitive, its pencil action is the cyclic regular prime-degree action of F^*/K^*, and its line stabilizer contains $K^+ \rtimes K^*$, sharply two-transitive on K. Its global action on F is sharply two-transitive as well. The local-action criterion proves every required property. This establishes existence within a one-dimensional group, without borrowing the full automorphism group of the affine space. □

Proposition 3.4 (Local two-transitivity in the one-dimensional branch)**.** *Under the spread hypotheses, local two-transitivity occurs only for $\mathrm{AG}(2,2)$ with $A\Gamma L_1(4)$, or $\mathrm{AG}(2,4)$ with $A\Gamma L_1(16)$.*

Proof. Write $v = p^{u^{e+1}}$, $k = p^{u^e}$ and $d = u^{e+1}$ by Theorem 3.3. Pencil two-transitivity forces $r(r-1) \mid d(v-1)$. Since $v - 1 = r(k-1)$, we have $r - 1 \mid d(k-1)$. But $k \mid r - 1$ and $\gcd(k, k-1) = 1$, so $k \mid d$. Hence $p = u$ and $u^e \leq e + 1$, leaving only $e = 0$ or $e = 1$; in the latter case $p = u = 2$. If $e = 0$, cancellation gives $1 + p + \cdots + p^{p-2} \mid p - 1$, which is possible only for $p = 2$. Thus the only parameter pairs are $(v, k) = (4, 2), (16, 4)$.

For four points, pencil two-transitivity requires a group order divisible by $4 \cdot 3 \cdot 2 = 24$, so the full $A\Gamma L_1(4)$ is necessary and is S_4. For sixteen points, the five-point pencil requires $16 \cdot 5 \cdot 4 \mid |G|$, and the twenty lines with four points each require $20 \cdot 4 \cdot 3 \mid |G|$. Their least common multiple

is 960, exactly $|A\Gamma L_1(16)|$; the full group is necessary. It suffices: on the five cosets $\mathbb{F}_{16}^*/\mathbb{F}_4^*$ the multipliers are a regular cyclic group of order five and Frobenius acts by multiplication by two, of order four on that quotient. This is sharply two-transitive. On a line $\mathbb{F}_4$, its translations and nonzero scalar multipliers give a sharply two-transitive group. □

4 Scalar quotients and the circle branches

Lemma 4.1 (A scalar-orbit test)**.** *Let F be a finite field and $C \leq F^*$. Suppose $F^+ \rtimes C \leq H \leq A\Gamma L_1(F)$. The affine action of H is two-transitive exactly when its zero stabilizer is transitive on F^*/C. If $\mathbb{F}_p[C] = F$, the action is primitive.*

Proof. Translations reduce ordered-pair transitivity to transitivity of the zero stabilizer on F^*. Scalar multiplication by C is transitive on each of its multiplicative cosets; these cosets are preserved as a system by all scalar-Frobenius maps. This proves the first equivalence. A block through zero is an additive subspace invariant under the zero stabilizer. If it contains a nonzero v and is invariant under C, it contains $\mathbb{F}_p[C]v = F$. Thus no proper nonzero block exists. □

Proposition 4.2 (Projective-line circles)**.** *Let $\mathrm{PSL}_2(q) \leq G \leq \mathrm{P\Gamma L}_2(q)$ in its natural degree-$q+1$ action. Its circle is always residually weakly primitive. It is locally two-transitive exactly when the induced action of G on the determinant square classes is transitive. For even q this condition is automatic; for odd q it says that some element has nonsquare determinant in its semilinear projective representative.*

Proof. The natural special-linear action is two-transitive. Fixing infinity identifies its remaining points with $\mathbb{F}_q$. Its induced stabilizer is the affine group $\mathbb{F}_q^+ \rtimes C$, where $C = (\mathbb{F}_q^*)^{\gcd(2,q-1)}$: the diagonal matrix $\mathrm{diag}(a, a^{-1})$ gives multiplication by a^2, and upper unipotents give every translation. General semilinear stabilizers give every allowed scalar-Frobenius map. The quotient character is determinant modulo squares, invariant under projective scalar changes. Since the special-linear group is transitive, the full point stabilizer maps onto the outer quotient. Lemma 4.1 therefore gives the claimed two-transitivity criterion on the remaining points, equivalent to three-transitivity on the whole set.

For primitivity, if q is prime, the additive group already has prime degree. Otherwise every proper subfield has size at most $\sqrt{q}$, whereas $|C| \geq (q-1)/2 > \sqrt{q}-1$. Thus C cannot be contained in a proper subfield and generates $\mathbb{F}_q$; Lemma 4.1 applies. Together with Proposition 1.4 this proves the assertion. The criterion retains mixed diagonal-field extensions; it does not incorrectly require the acting group to contain $\mathrm{PGL}_2(q)$. □

Lemma 4.3 (A characteristic subgroup in a regular normal subgroup)**.** *If a transitive group on X contains a regular normal subgroup U with a nontrivial proper characteristic subgroup Z, then it is imprimitive.*

Proof. The Z-orbits are blocks, since Z is normal in the full group. Regularity identifies their size with $|Z|$, strictly between one and $|X| = |U|$. □

This proves the local exclusion for the natural Suzuki and Ree circle actions from their precise retained rank-one structure: a point stabilizer has a regular unipotent radical of order q^2 or q^3, respectively, and a proper nontrivial characteristic center. Field and torus automorphisms preserve this center. Thus the remaining-point action of the point stabilizer is imprimitive, also for its allowed outer extensions. No assertion that a Suzuki ovoid is itself a thick linear space is needed. Natural unitary circles instead preserve the actual thick unital, so Proposition 1.4 directly excludes them. Projective actions of dimension at least two preserve their thick projective lines and are excluded in the same way.

For affine circles over an odd prime field, the actual affine lines are thick and again give an invariant pencil partition. For extension field actions on an even field larger than two, the same

exclusion uses the actual invariant extension-field lines, after establishing their normalizer as in Lemma 1.3. Over $\mathbb{F}_2$, by contrast, affine lines have only two points and cannot be used for this exclusion. The natural special-linear family gives three-transitivity, and the symplectic family gives primitivity but not two-transitivity on the nonzero vectors by Proposition 2.2. The exceptional $G_2(2)'$ degree-63 local action requires its own small case: it must not be excluded by pretending $\mathrm{AG}(6,2)$ is thick. The following Proposition 4.4 handles it using the natural module as a retained input and a complete small maximal-subgroup calculation; the other exceptional action interfaces are kept separate.

The one-dimensional circle branch is covered by Theorem 3.3 with $\ell = 1$: the pencil size $p^d - 1$ must be prime. This includes, for example, $\mathrm{AGL}_1(8)$ with primitive cyclic pencil of degree seven. It is omitted from the literal stand-alone list in source Proposition 4.3, whose proof defers the one-dimensional case to Section 5. The source's final Theorem 6.1 does include it in its first branch. The corresponding locally two-transitive circle is only the four-point case among one-dimensional spread actions.

For the exceptional almost-simple branches we will retain exact permutation-action information, rather than infer primitivity from abstract group names or orders alone. In particular the A_7 action on fifteen points preserves the lines of $\mathrm{PG}(3,2)$. It is two-transitive on points but its point stabilizer preserves seven pairs on the other fourteen points. Its circle is not locally primitive and cannot be three-transitive. This is the previously accepted Topic G circle correction, not a new conjecture result.

Proposition 4.4 (The binary exceptional local action)**.** *Retain the natural transitive binary representation of dimension six of $N = G_2(2)' \cong \mathrm{PSU}_3(3)$. Every group $N \le H \le N_{\mathrm{GL}_6(2)}(N)$ is primitive but not two-transitive on the sixty-three nonzero vectors. Thus $2^6 \rtimes H$ has a residually weakly primitive circle action and is not locally two-transitive.*

Proof. The small exact finite premise supplied by `small_g2_premise.g` is the complete maximal-subgroup order list

$$|N| = 6048, \qquad 96, 96, 168, 216, \qquad |\operatorname{Aut}(N)| = 12096.$$

It is computed in GAP from $\mathrm{PSU}(3,3)$, including a simplicity check and both actual degree-63 coset actions. This is a required finite premise, not an assertion derived solely from a subgroup order. In the natural transitive action a vector stabilizer has order $6048/63 = 96$. Any proper overgroup of it lies in a maximal subgroup, and no order in the displayed complete list is a larger multiple of 96. Thus the stabilizer is maximal and the action is primitive.

For the normalizer bound we can avoid assuming an unverified full matrix normalizer. Transitivity on nonzero vectors implies irreducibility. Its commuting algebra is a finite division ring, hence a field $\mathbb{F}_{2^e}$, with $e \mid 6$. If $e = 3$ or 6, Lagrange's theorem would embed N in $\mathrm{GL}_2(8)$ or $\mathrm{GL}_1(64)$, whose orders are not divisible by 6048. If $e = 2$, simplicity and perfectness put N inside $\mathrm{SL}_3(4)$; its intersection with the central subgroup of order three is trivial, so it embeds in $\mathrm{PSL}_3(4)$, of order 20160, again not divisible by 6048. Consequently $e = 1$, and its invertible commuting algebra is trivial. Conjugation embeds the entire linear normalizer in $\mathrm{Aut}(N)$. A two-transitive action on sixty-three points would require order divisible by $63 \cdot 62$, which does not divide 12096. Every intermediate group inherits primitivity from N, but none can be two-transitive. Proposition 1.4 proves the conclusion. □

Proposition 4.5 (Orthogonal-point symplectic actions)**.** *Retain the standard two-transitive actions of $\mathrm{Sp}_{2n}(2)$ of degrees $2^{2n-1} \pm 2^{n-1}$, $n \ge 3$, and the classical rank-three subdegrees of their orthogonal point stabilizers. Both circle actions are locally primitive and neither is locally two-transitive.*

Proof. Put $b = 2^{2n-2}$ and $d = 2^{2n-1} \pm 2^{n-1} - 1$. On the d remaining points, a point stabilizer of the orthogonal group has orbits of lengths 1, $a = d - 1 - b$, and b; these are the precise retained

rank-three data. A block containing the fixed point is a union of these orbits, so its only possible proper nontrivial sizes are $a+1 = d-b$ and $b+1$. The first is an odd integer greater than one; if it divided d, it would divide the power of two b, impossible. For the minus sign, $d/2 < b+1 < d$. For the plus sign, $2(b+1) < d < 3(b+1)$, since $n \geq 3$. Thus $b+1$ cannot divide d either. The local action is primitive. Its rank is three, so it is not two-transitive. The circle criterion proves both conclusions. □

5 The sharply flag-transitive and nonspread branches

We retain the projective flag-transitivity classification in source Proposition 3.1, attributed to Kantor and Cameron–Kantor: besides the classical group and the A_7 action on $\mathrm{PG}(3,2)$, a flag-transitive projective plane has prime point number m^2+m+1 and a sharply flag-transitive Frobenius group of order $(m^2+m+1)(m+1)$. We also retain Feit's restrictions in Proposition 3.2: the Desarguesian possibilities are $m = 2, 8$, while another example requires $m \equiv 0 \pmod 8$ and $d^{m+1} \equiv 1 \pmod{m^2+m+1}$ for every divisor d of m. The reported computer extension of Feit's bound to $m \leq 10^{10}$ is not a premise of the following proof and has no supplied source program.

Proposition 5.1 (The exceptional projective branch)**.** *A locally primitive sharply flag-transitive projective plane in this retained branch is the Fano plane with group* $7:3$*. It is not locally two-transitive.*

Proof. The point stabilizer acts regularly on its $m+1$ incident lines. A regular action is primitive only in prime degree, so $m+1$ is prime. For a putative non-Desarguesian example, put $N = m^2+m+1$. We have $m^3 \equiv 1 \pmod N$, and the order of m modulo the prime N is exactly three, since $1 < m < N$. Feit's condition with $d = m$ therefore gives $3 \mid m+1$, forcing $m = 2$, a contradiction. Among the Desarguesian possibilities, only $m = 2$ has $m+1$ prime. The group $7:3$ is regular on flags of the Fano plane; both local three-point groups are cyclic regular. They are primitive but are not two-transitive. □

The other exceptional projective action has A_7 on fifteen points and thirty-five lines of $\mathrm{PG}(3,2)$. Retain its classical identification and point stabilizer $\mathrm{PSL}_3(2)$ from Cameron–Kantor/Taylor, as in source Proposition 3.3. The pencil is its natural seven-point plane action, two-transitive by the elementary basis argument. The stabilizer of a line induces S_3 on its three points. With these exact retained local identifications, its claimed local two-transitivity follows. This geometric action must not be confused with the circle on the same fifteen points.

Proposition 5.2 (One-dimensional nonspread actions, including the small boundary)**.** *Retain Delandtsheer's translation/nonspread reduction quoted in source Propositions 5.2 and 5.7. A flag-transitive one-dimensional nonspread linear space is locally primitive precisely in the following two cases: the three-point circle with* $S_3 = \mathrm{AGL}_1(3)$*, and the Fano plane with* $7:3$*. Only the first is locally two-transitive.*

Proof. Write $v = p^d$ and let k and $r = (v-1)/(k-1)$ be line size and pencil size. The point stabilizer is a subgroup of $\Gamma L_1(p^d)$, cyclic by cyclic. Its transitive action is primitive exactly when r is prime, by Lemma 3.1 and the elementary fact that a transitive prime-degree action is primitive. The retained nonspread reduction gives $v \mid b$, where $b = vr/k$, and hence $k \mid r$. Thus $k = r$ and $v = k(k-1)+1$.

There are now $b = v$ lines. Every other line meets a fixed line L: its k points lie on $k-1$ other lines each, giving $k(k-1) = v-1$ distinct lines, since two distinct points of L cannot share another line. If $k = 2$ this is exactly the three-point circle. Flag transitivity requires all six permutations, so the specified group is S_3; both local actions have degree two.

If $k \geq 3$, this is a projective plane of order $k-1$. The retained projective classification and the preceding proposition leave exactly the Fano plane and its group $7:3$. Conversely these two structures have the asserted one-dimensional realizations and primitive local actions. The cyclic local groups in the Fano case are not two-transitive. □

The source's Proposition 5.8 omits the $k = 2$ case when passing from $r = k$ to a nondegenerate projective plane. Its earlier Proposition 5.1 already lists this three-point circle. The footnote after Proposition 5.6 saying the spread restriction can be removed must therefore include this extra three-point action. The final circle branches of Theorems 6.1 and 6.2 already include it as the natural degree-three $\mathrm{PSL}_2(2)$ action. No new family beyond that final union is being claimed.

6 Comparing different affine structures

The printed proof of source Lemma 3.5 assumes that two invariant affine structures are nested merely because their line sizes are ordered. The following comparison does not make that assumption.

Lemma 6.1 (An intrinsic translation group)**.** *Let G act primitively on a set X of size p^d, $d \geq 2$, and contain a regular elementary abelian normal subgroup V. If G preserves a Desarguesian affine-space structure on X, then V is its translation group.*

Proof. The point stabilizer acts irreducibly on V: a proper invariant subgroup would give a nontrivial block system of its cosets. Thus V is minimal normal in G. In the given affine structure, project G to its cyclic field-automorphism group. The kernel of the restriction to V is normal in G and is nontrivial, because an elementary abelian group of rank at least two cannot inject into a cyclic group. Minimality puts all of V in the affine linear subgroup.

Take linear parts there. A finite p-subgroup of a linear group in characteristic p has a nonzero fixed vector: orbit counting on the vector space makes the number of fixed vectors divisible by p. The common fixed space W of the image of V is invariant under the entire semilinear image of G, since $V \lhd G$. If W were proper, its affine cosets would be a nontrivial G-invariant partition of X. Thus W is the whole vector space. The linear image of V is trivial, so V consists of translations; equality follows from the common order p^d. □

Proposition 6.2 (Scalar rigidity)**.** *In Lemma 6.1, identify X with V by choosing an origin. Suppose its stabilizer contains a perfect subgroup N and $\mathrm{End}_{\mathbb{F}_p N}(V) = \mathbb{F}_q$. Then every G-invariant Desarguesian affine structure has line size at most q. Its scalar field is an actual subfield of this commuting algebra. If the natural $\mathbb{F}_q$-structure is G-invariant, it has largest line size among these structures.*

Proof. All structures have the same translation group by the lemma, hence the same addition on V. In another structure over $\mathbb{F}_Q$, the field action of N is a homomorphism to a cyclic group and is trivial because N is perfect. Thus N is $\mathbb{F}_Q$-linear. Every scalar of $\mathbb{F}_Q$ commutes with N, giving an embedding of that actual scalar field into $\mathrm{End}_{\mathbb{F}_p N}(V) = \mathbb{F}_q$. In particular $Q \leq q$. This argument constructs the common scalar setting rather than assuming a prior partition of one pencil by the other. □

For natural special-linear modules of dimension at least three and natural symplectic modules of dimension at least four in the thick range $q \geq 3$, the perfectness and the commuting-algebra calculations above apply. For the even-field natural $G_2(q)$ module, the commuting algebra follows from an order argument. Retain its standard faithful six-dimensional representation, transitivity on nonzero vectors, simplicity for $q \geq 4$, and order $q^6(q^6-1)(q^2-1)$. Transitivity gives irreducibility over $\mathbb{F}_p$. Its commuting algebra is a finite division ring, hence a field containing $\mathbb{F}_q$, of relative degree dividing six. Relative degrees three or six would embed the group into $\mathrm{GL}_2(q^3)$ or $\mathrm{GL}_1(q^6)$, impossible by the p-parts of their orders. Relative degree two would embed this perfect simple group into $\mathrm{SL}_3(q^2)$ and then, with trivial central intersection, into $\mathrm{PSL}_3(q^2)$. The ratio of the latter order to $q^6(q^6-1)(q^2-1)$ is $(q^2+1)/3$, not an integer since q is a power of two. Thus the commuting algebra is exactly $\mathbb{F}_q$. The $q = 2$ derived-group case has been proved separately in Proposition 4.4.

For the one-dimensional branch no comparison of affine structures is needed to bound line size. A primitive pencil has prime size by Lemma 3.1; if its line size is p^f and point size p^d, that lemma gives $d = u^{a+1}$ and $f = u^a$ for a prime u. Every nondegenerate affine space on p^d points has line size p^b where b is a proper divisor of d, so $b \le u^a = f$. This proves maximality even against non-Desarguesian affine planes.

Proposition 6.3 (The natural even-field exceptional pencil)**.** *Retain the classical six-dimensional symplectic representation of $G_2(q)$, $q \ge 4$ even, in which the points of the split Cayley hexagon are exactly the vector lines, and their stabilizers are maximal parabolic subgroups. Every group between this group and its full $\mathbb{F}_2$-linear normalizer is primitive but not two-transitive on vector lines. With translations its affine action is locally primitive but not locally two-transitive. The analogous binary derived-group assertion is Proposition 4.4.*

Proof. Transitivity and maximality of the stated point stabilizers imply primitivity; these are the exact retained building inputs, attributed to the classical hexagon identification used in source Proposition 3.6. The commuting-field calculation above makes every normalizer element semilinear over $\mathbb{F}_q$. Remove its field automorphism and pull back the preserved nondegenerate alternating form. The resulting form is again $G_2(q)$-invariant. Its defining operator relative to the original form commutes with $G_2(q)$, so it is a scalar by the calculated commuting algebra. Hence the entire normalizer preserves perpendicularity on vector lines. There are distinct perpendicular and nonperpendicular pairs in dimension six, so none of these groups is two-transitive on the pencil. The retained transitivity of the natural group on nonzero vectors implies transitivity of a line stabilizer on its nonzero vectors. Translations then give two-transitivity on each affine line. Proposition 1.2 proves the remaining assertions. □

Lemma 6.4 (Large vector-stabilizer orbits)**.** *Let $V = \mathbb{F}_q^n$, $q \ge 3$. If $n \ge 3$ and $N = \mathrm{SL}_n(q)$, then N_v, for $0 \ne v \in V$, is transitive on $V \setminus \mathbb{F}_q v$, of size $q^n - q > q^{n/2}$. If $n \ge 4$ is even and $N = \mathrm{Sp}_n(q)$, every N_v-orbit outside $\mathbb{F}_q v$ has size at least $q^{n-1} - q > q^{n/2}$. Consequently a G-invariant affine-space structure on V, where $G_0 \ge N$, has no line larger than q.*

Proof. For the special-linear assertion, extend (v, w) and (v, w') to bases for any $w, w' \notin \mathbb{F}_q v$. Map the bases to each other and correct the determinant on a third basis vector. This fixes v and sends w to w'. For the symplectic assertion, symplectic basis extension gives one orbit $v^\perp \setminus \mathbb{F}_q v$, of size $q^{n-1} - q$, and one orbit $\{w : \langle v, w\rangle = c\}$, of size q^{n-1}, for each $c \ne 0$. The first transitivity follows by extending an ordered independent isotropic pair; the second follows by extending the hyperbolic pair $(v, c^{-1}w)$. These constructions account for every vector outside $\mathbb{F}_q v$. The displayed strict inequalities hold already at $n = 3$ and $n = 4$, respectively, and increase thereafter.

Here the comparison is only with affine spaces, not arbitrary linear spaces. Every nondegenerate finite affine space on $|V| = q^n$ points has line size at most $\sqrt{|V|}$: a plane has exactly k^2 points if its line size is k, and higher dimension gives at least a plane. For an alternative line L through 0 and v, uniqueness of the line through these two points makes L invariant under N_v. It cannot contain any of the above outside orbits. Thus $L \subseteq \mathbb{F}_q v$, and its size is at most q. □

Theorem 6.5 (Maximal line size for a primitive affine pencil)**.** *Let a fixed faithful permutation group G on a finite set X act flag-transitively on an affine space Γ with at least two lines through each point. If G_x is primitive on its pencil, then Γ has largest line size among all G-invariant affine-space structures on X on which G is flag-transitive. This recovers source Lemma 3.5 by contraposition. Its proof retains the precise BDDKLS–Huybrechts affine classification and the BDD exclusion of local point-primitivity for the full Lüneburg-plane group.*

Proof. A plane already has line size $\sqrt{|X|}$, the universal affine bound just proved. If the lines of Γ have size two, a larger-line structure would partition each pencil of the circle into the sets $L' \setminus \{x\}$, contradicting Proposition 1.4. We may therefore suppose $\Gamma = \mathrm{AG}(n, q)$, $n \ge 3$, $q \ge 3$; the coordinatization of affine spaces of dimension at least three is the classical retained input.

Use exactly the classification interface quoted in source Section 3.2.1: a flag-transitive group is one-dimensional, belongs to the excluded $\mathrm{AG}(4,3)$ last-derived-$2.A_5$ case, or is two-transitive with one of Huybrechts' six stated natural representation types. The excluded case is a retained impossibility, not inferred here from its abstract group name. In the one-dimensional case the prime-pencil and prime-power argument immediately preceding this theorem proves the line-size bound, including comparison with non-Desarguesian planes.

In the two-transitive cases write $G = V \rtimes G_0$. The three small exceptional representation types have imprimitive pencils by the explicit exclusions in the classical-pencils section. For a natural special-linear, symplectic or even-field G_2 normal subgroup over $\mathbb{F}_{q^t}$, the commuting-algebra arguments make G_0 normalize that actual scalar field. If $t > 1$, its lines partition the $\mathbb{F}_q$-pencil nontrivially, contrary to Lemma 1.3. (The special-linear one-dimensional case belongs to the one-dimensional alternative.) Thus the remaining modules are natural over $\mathbb{F}_q$ itself. In the special-linear and symplectic cases Lemma 6.4 applies also to non-Desarguesian alternative planes and proves the conclusion.

It remains to consider $N = G_2(q)$ on $V = \mathbb{F}_q^6$, with q even and $q \geq 4$. This subgroup is perfect and its commuting field is $\mathbb{F}_q$, as proved above. Scalar rigidity bounds every alternative Desarguesian line size by q. Any remaining alternative is a non-Desarguesian affine plane. It cannot have a one-dimensional flag-transitive group: every subgroup of $A\Gamma L_1(p^d)$ is soluble, whereas G contains the nonabelian simple subgroup N. The retained BDD list of non-one-dimensional flag-transitive non-Desarguesian planes consists of the Lüneburg planes, the nearfield plane of order nine and the Hering plane of order twenty-seven. The latter two have an odd number of points, whereas $|V| = q^6$ is even. Thus only a Lüneburg plane could remain.

But G is two-transitive on V, independently of its incidence structure. In any invariant linear space this implies two-transitivity of every line stabilizer on that line: an element sending one ordered pair of distinct points of the line to another sends their unique joining line to itself. Hence that line action is primitive, and remains primitive in the full automorphism group by Lemma 1.1. This contradicts the retained BDD fact, quoted in source Section 3.2.2, that the full Lüneburg-plane group is not locally point-primitive. This eliminates the last alternative and completes the proof. □

The argument does not assert that two arbitrary invariant affine structures are nested. Its orbit bounds and classification comparison replace the unproved nesting sentence and the unnecessary full-group equality in the printed proof. The classification inputs above remain charged retained inputs; their proofs have not been reconstructed here.

7 The exact scalar obstruction in a Hermitian pencil

Let $q = p^f \geq 2$, $F = \mathbb{F}_{q^2}$, and put $d = \gcd(3, q+1)$. The classical unital has the isotropic projective points for

$$h(X) = X_0^q X_2 + X_0 X_2^q + X_1^{q+1}.$$

Its lines are the secants, with incidence inherited from the plane. The standard full projective semilinear unitary group and its special subgroup are denoted by $\mathrm{P\Gamma U}_3(q)$ and $\mathrm{PSU}_3(q)$. We retain the classical identification of the full unital automorphism group with $\mathrm{P\Gamma U}_3(q)$. The local computations below are explicit and do not retain the blanket local-two-transitivity assertion of the source.

Let $C = (F^*)^d$ and $\mathcal{C} = F^*/C$. There is a natural action

$$\pi : \mathrm{P\Gamma U}_3(q)/\mathrm{PSU}_3(q) \longrightarrow \mathrm{Sym}(\mathcal{C}). \tag{U1}$$

To specify it, fix $\infty = \langle (1,0,0) \rangle$ and identify its pencil with F. A point stabilizer acts as $b \mapsto ab^{p^i} + c$; send this to $Cz \mapsto Caz^{p^i}$. Changing a representative by a special-unitary stabilizer multiplies a by C and changes only the translation c. Because the special-unitary group is point-transitive, the

full group's point stabilizer surjects onto the full outer quotient. This defines (U1). When $d = 3$, choosing a generator of F^*/C identifies the action with $j \mapsto t + p^i j$ on $\mathbb{Z}/3\mathbb{Z}$.

Theorem 7.1 (Unitary local criterion)**.** *Suppose* $\mathrm{PSU}_3(q) \le G \le \mathrm{P\Gamma U}_3(q)$ *in its specified unital action. Then G is flag-transitive and its line stabilizers are two-transitive on their incident points. The point stabilizers are two-transitive on their incident lines if and only if $\pi(G)$ is transitive on $\mathcal{C}$. For $q > 2$ the action is always residually weakly primitive. For $q = 2$ it is residually weakly primitive if and only if the same transitivity condition holds; in that case it is locally two-transitive.*

Proof. Every unital point other than ∞ has a unique representative $(a, b, 1)$ with $a + a^q + b^{q+1} = 0$. Its joining line with ∞ depends only on b. Consequently the pencil at ∞ identifies with F. For $c, t \in F$ satisfying $t + t^q = -c^{q+1}$, the matrices

$$u(c,t) = \begin{pmatrix} 1 & -c^q & t \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \tag{U2}$$

preserve h and have determinant one. Their action on the pencil is $b \mapsto b + c$. The induced translation group therefore is all F^+. The determinant-one diagonal matrices

$$\mathrm{diag}(z, z^{q-1}, z^{-q}), \quad z \in F^*, \tag{U3}$$

induce $b \mapsto z^{2q-1}b$. The image of this power map is C, because $\gcd(2q-1, q^2-1) = \gcd(3, q+1) = d$.

Conversely, a projective unitary element fixing ∞ preserves its orthogonal tangent. In the displayed basis its matrix is upper triangular. The unitary equations give (U2) followed by a unitary diagonal matrix; the determinant-one condition gives (U3), up to a projective scalar. Thus the induced special-unitary pencil group is exactly $F^+ \rtimes C$, not merely a subgroup of that group. General projective unitary diagonals give all multipliers in F^*, and field automorphisms give $b \mapsto b^{p^i}$. This also verifies the outer action (U1).

The full group G_∞ contains every translation. Removing a translation from each element fixing an ordered pair reduces pencil two-transitivity to transitivity on F^* of its zero stabilizer. That stabilizer contains C, which acts regularly on each multiplicative coset of C. It is therefore transitive on F^* exactly when the induced action on $\mathcal{C}$ is transitive. This proves the claimed criterion.

For primitivity, C generates the field F over $\mathbb{F}_p$ when $q > 2$. Indeed $|C| = (q^2 - 1)/d > q - 1$, whereas the multiplicative group of any proper subfield of F has order at most $q - 1$. Thus no nonzero proper additive subspace can be invariant under C. The affine pencil action is primitive by Proposition 1.2's block argument. When $q = 2$, $C = 1$, and the zero stabilizer lies in $\Gamma L_1(4) = \mathrm{GL}_2(2)$. Its action on the four-point additive space is irreducible exactly when it contains an element of order three, equivalently when it is transitive on the three nonzero vectors. This is precisely the criterion on $\mathcal{C}$ in this case.

For completeness the line action is two-transitive already in the special unitary subgroup. A secant spans a nondegenerate two-dimensional Hermitian space and its orthogonal complement is a nonsingular point. The subgroup acting as $\mathrm{SU}_2(q)$ on the former and identically on the latter lies in $\mathrm{SU}_3(q)$. In a hyperbolic basis the isotropic points of the former are ∞ and $\langle (t, 1) \rangle$ with $t + t^q = 0$. The matrices $\left(\begin{smallmatrix} 1 & c \\ 0 & 1 \end{smallmatrix}\right)$, $c + c^q = 0$, fix ∞ and act regularly on the other q points. For any nonzero such c, the determinant-one isometry $\left(\begin{smallmatrix} 0 & c \\ -c^{-1} & 0 \end{smallmatrix}\right)$ interchanges ∞ and zero. Hence this group is two-transitive on all $q + 1$ points.

The same rank-one isometries give point-transitivity of $\mathrm{PSU}_3(q)$, and (U2) gives transitivity on each pencil. Thus it is flag-transitive. Global point-primitivity follows, for example, from its usual two-transitive unital action: (U2) is regular on the q^3 other unital points. Together with the pencil and line calculations this proves every assertion. □

Corollary 7.2 (Two literal source boundaries)**.** *For $d = 1$, every group between* $\mathrm{PSU}_3(q)$ *and* $\mathrm{P\Gamma U}_3(q)$ *is locally two-transitive. For $d = 3$,* $\mathrm{PSU}_3(q)$ *itself is not locally two-transitive. In particular* $\mathrm{PSU}_3(2)$ *is not residually weakly primitive on the unital* $\mathrm{AG}(2, 3)$*, although* $\mathrm{PGU}_3(2)$ *is.*

Proof. If $d = 1$ there is one coset. If $d = 3$, the special-unitary image in (U1) is trivial on three cosets. The last assertion uses the $q = 2$ part of Theorem 7.1; the projective unitary image supplies the cyclic regular action on those three cosets. As a small order check, $|\mathrm{PSU}_3(2)| = 72$ and its point stabilizer has order eight, so it cannot be two-transitive on the four incident lines. □

8 Witt–Bose–Shrikhande local actions

We use the standard nonsplit-torus construction of the Witt–Bose–Shrikhande space, attributed in source Section 3.4 to the earlier work cited there. Its full automorphism group $\mathrm{P\Gamma L}_2(q)$ is the retained classification input from BDD, as used by the source; the local-action argument below is given in full. In particular, the full-group identification is not inferred from an exhibited subgroup. Since q is even, $\mathrm{PSL}_2(q) = \mathrm{PGL}_2(q) = \mathrm{SL}_2(q)$.

Proposition 8.1 (Torus–involution incidence)**.** *Let $q = 2^e$, $e \geq 3$, and $G = \mathrm{SL}_2(q)$. Let the points be normalizers $D = N_G(T)$ of nonsplit tori T of order $q + 1$, and let the lines be the involutions s of G. Declare D incident with s when $s \in D$. This is a flag-transitive linear space with*

$$v = q(q-1)/2, \qquad b = q^2 - 1, \qquad k = q/2, \qquad r = q + 1.$$

A point stabilizer in G is D of order $2(q+1)$, a line stabilizer is $U = C_G(s)$ of order q, and a flag stabilizer is $\langle s \rangle$.

Proof. Identify the two-dimensional vector space with $\mathbb{F}_{q^2}$. Multiplication by its norm-one elements gives the cyclic group T. The map $z \mapsto z^q$ inverts T and, in characteristic two, has determinant one. The normalizer is therefore the dihedral group D of order $2(q+1)$: its elements either fix or exchange the two common eigenlines over $\mathbb{F}_{q^2}$. Conversely these are all possibilities for a normalizer. Every nonidentity element of T has distinct eigenvalues outside $\mathbb{F}_q$, since $\gcd(q-1, q+1) = 1$, and its centralizer in G is exactly T. All nonsplit tori are conjugate: conjugate their two-dimensional field embeddings, and adjust a conjugating matrix's determinant by a centralizing field multiplier, whose norm ranges over $\mathbb{F}_q^\times$. Also D is self-normalizing, since its odd cyclic subgroup is characteristic. This proves the point count.

Every involution is conjugate to $s = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Its centralizer consists exactly of $u_a = \left(\begin{smallmatrix} 1 & a \\ 0 & 1 \end{smallmatrix}\right)$, $a \in \mathbb{F}_q$, as direct matrix multiplication shows. Thus U is elementary abelian of order q and there are $q^2 - 1$ involutions. The $q+1$ reflections of D form one T-orbit, because squaring is bijective on T. Consequently G is flag-transitive and the flag stabilizer is $C_D(s) = \langle s \rangle$. This proves $r = q + 1$ and $k = q/2$.

Two distinct point normalizers cannot contain two distinct common involutions: their nontrivial product lies in the rotation torus of each, and its unique centralizer identifies those tori and hence the normalizers. Thus two points lie on at most one line. Moreover

$$b\binom{k}{2} = \frac{(q^2-1)q(q-2)}{8} = \binom{q(q-1)/2}{2}.$$

Every pair therefore lies on exactly one line. The parameter bounds give at least two points on every line and two lines on every point. □

Theorem 8.2 (Uniform local exclusion and a corrected source sentence)**.** *For $e \geq 3$, no flag-transitive subgroup of the full group $A = \mathrm{P\Gamma L}_2(2^e)$ of $W(2^e)$ is residually weakly primitive. Nevertheless the full group of $W(8)$ is primitive on the points of each line: its induced line action is $\mathrm{AGL}_1(4) \cong A_4$. Thus the final WBS exclusion of source Section 3.4 is valid, but its stronger assertion that the full group is never locally point-primitive is false.*

Proof. First consider the pencil at D. Its lines are the $q + 1$ involutions in D. The rotation subgroup T acts regularly on them. It is normal in A_D, since it is characteristic in D and $G \lhd A$. If $q + 1$ is composite, a proper nontrivial characteristic subgroup of the cyclic T gives a nontrivial

A_D-invariant partition into its orbits. Thus pencil primitivity forces $2^e + 1$ to be prime. Writing $e = 2^a m$ with m odd shows that $m = 1$: for $m > 1$, $2^{2^a} + 1$ is a proper divisor. Hence e is a power of two, and $e \geq 4$ in the present range.

Now consider the line represented by the displayed involution $s = u_1$. Its stabilizer is the centralizer A_s. Each element of A is a matrix part followed by a field automorphism. Field automorphisms fix s; the matrix centralizer calculation above gives

$$A_s = U \rtimes \langle \sigma \rangle, \qquad \sigma(a) = a^2, \qquad |\langle \sigma \rangle| = e.$$

Fix a point D on this line and put $B = A_s \cap A_D$. Since G is flag-transitive, every coset of G in A has a representative fixing this flag. Hence $B/(B \cap G) \cong A/G \cong C_e$, whereas $B \cap U = B \cap G = \langle s \rangle$. Also $A_s = UB$. The line points can consequently be identified with the additive quotient

$$V = U/\langle s \rangle \cong \mathbb{F}_{2^e}/\mathbb{F}_2,$$

with V acting regularly. Conjugation by the point stabilizer acts on this quotient through $\langle \sigma \rangle$; translation parts have no effect on that action. The affine block criterion in Proposition 1.2 says that line primitivity is equivalent to irreducibility of this stabilizer on V.

For the only remaining pencil possibility, e is a power of two. The stabilizer image on V is therefore a 2-group. A 2-group acting on a nonzero vector space over $\mathbb{F}_2$ has a nonzero fixed vector: all its nonfixed orbits have even size, so the number of fixed vectors is even and includes zero. Since $\dim_{\mathbb{F}_2} V = e - 1 > 1$, a fixed one-dimensional subspace is a proper invariant subspace. The line action is imprimitive. Thus the full group fails at least one of the two required local actions for every e. A primitive local action of a subgroup would make the corresponding full group action primitive by Lemma 1.1; no subgroup can evade this obstruction. This proves the uniform exclusion.

For the stronger source assertion, set $e = 3$. On $V = \mathbb{F}_8/\mathbb{F}_2$, a class fixed by σ has $x^2 + x \in \mathbb{F}_2$. The value zero gives the zero class; the value one would put a root of $X^2 + X + 1$ in $\mathbb{F}_8$, impossible because its roots have multiplicative order three whereas $|\mathbb{F}_8^\times| = 7$. The order-three action on the two-dimensional V is therefore irreducible. It is transitive on the three nonzero vectors. The induced line group is $V \rtimes C_3$ of order twelve, proving the claimed primitive action. Its pencil remains imprimitive since $q + 1 = 9$. The short accompanying $q = 8$ calculation validates these exact stabilizers but is not a premise of the written proof. □

9 Dependency and review boundary

The companion application supplies the full source dispositions and both corrected classification converses. All proofs in this common paper are written here; no earlier common theory is treated as a free imported proof. The semilinear spread descent is reused from the already reviewed phase-four argument with its prior scope unchanged. The large classification inputs and classical representation identifications are retained at their named point of use. The binary exceptional pencil uses a complete small finite premise, whereas the WBS diagnostic only validates a uniform written proof. No new Lean declaration or whole-source admission is claimed by this private unreviewed edition.

References

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